ξ-Normalized BAO Residuals in DESI DR1 and DR2: A Matter-Density Deficit and a Provisional D_H/r_s Signal

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Abstract

The reported 5σ Hubble tension is widely interpreted as a breakdown of the ΛCDM model. By applying Horizon-Normalized Coordinates ($\xi = d_c/d_H$), we find evidence that the tension decomposes into separable layers rather than a single monolithic discrepancy. We confirm that $\sim$90% of the apparent tension is a "frame-mixing" artifact (Martin, 2025b). The physical residual partitions further into (at minimum) two additional layers. Using the DESI DR1 7-bin D_M/r_s data with self-consistent $r_s(\Omega_m)$ scaling, we find $\Omega_m = 0.268$ ($\Delta\chi^2_{\text{Planck}} = 7.06$; $\Delta\chi^2_{w_0w_a} = 0.12$, statistically unsupported). Extending to the full DESI DR2 consensus dataset (DESI Collaboration, 2025)—13 measurements of D_M/r_s and D_H/r_s with the complete covariance matrix—the Ω_m deficit is supported at $\Omega_m = 0.290$, consistent with the DR2 headline value of 0.289 ± 0.007. A third residual pattern appears in the D_H/r_s data: adding w_0w_a freedom yields $\Delta\chi^2 = 13.13$ ($\chi^2/\text{dof} = 0.79$), driven by D_H/r_s residuals at $z = 0.51$ (-1.84σ) and $z = 0.71$ (-2.21σ below ΛCDM). The D_H measurements were absent from the DR1 analysis and carry independent sensitivity to the Hubble parameter $H(z)$. Within this framework, the tension decomposes into three layers: (1) $\sim$90% coordinate artifact, (2) a supported Ω_m deficit, and (3) a provisional D_H-driven line-of-sight residual pattern visible only in DR2. Independent confirmation of all three layers awaits Euclid DR1.

1 Introduction

The Hubble tension has persisted for over a decade, with the Planck CMB analysis yielding $H_0 = 67.4 \pm 0.5$ km/s/Mpc (Planck Collaboration, 2020) and the SH0ES distance ladder yielding $H_0 = 73.04 \pm 1.04$ km/s/Mpc (Riess et al., 2022). The 5σ discrepancy has motivated proposals for early dark energy (Poulin et al., 2019), modified gravity, and non-standard recombination. In Martin (2025b), we demonstrated that 93.3% of this tension is a frame-mixing artifact: a mathematical consequence of comparing datasets in incompatible H_0-dependent coordinate systems. The residual physical signal is a $\sim$6% matter density discrepancy.

Here we extend that analysis to characterise the residual using both the DESI DR1 and DR2 BAO datasets (DESI Collaboration, 2024a,b, 2025). The DR1 D_M/r_s-only

analysis (7 bins) confirms an Ω_m deficit with $\Delta\chi^2_{w_0w_a} = 0.12$ (statistically negligible). The DR2 consensus dataset adds D_H/r_s measurements at each redshift, enabling direct sensitivity to $H(z)$. We find that the D_H/r_s data carry an independent dark energy signal ($\Delta\chi^2 = 13.13$) invisible in the D_M-only analysis. The data suggest a three-layer structure in the tension residuals.

2 Method: ξ-Normalization Applied to BAO

We apply the horizon-normalized coordinate framework from Martin (2025b):

$$\xi(z) = \frac{d_c(z)}{d_H} = \int_0^z \frac{dz'}{E(z')} \tag{1}$$

where for the general w_0w_a dark energy parametrization (CPL; Chevallier & Polarski 2001; Linder 2003):

$$E(z) = \sqrt{\Omega_m(1+z)^3 + (1-\Omega_m)(1+z)^{3(1+w_0+w_a)}e^{-3w_az/(1+z)}} \tag{2}$$

For redshift-derived distances, $d_c \propto 1/H_0$ and $d_H \propto 1/H_0$, so ξ is exactly H_0-independent. We apply this to all 7 DESI DR1 BAO bins spanning $0.295 \leq z \leq 2.33$ (DESI Collaboration, 2024a).

A critical methodological requirement is that the sound horizon r_s must be varied self-consistently with Ω_m. Fixing r_s at the Planck fiducial (147.09 Mpc) while varying Ω_m inverts the sign of the fit, producing a spurious apparent excess rather than the physical deficit. We use the Percival et al. 2007 scaling:

$$r_s(\Omega_m) = 147.09 \left(\frac{\Omega_m h^2}{0.1430}\right)^{-0.255} \text{Mpc} \tag{3}$$

normalized to the Planck 2018 value at $\Omega_m = 0.315$, $h = 0.674$ (Planck Collaboration, 2020). This ensures D_M/r_s is computed with both numerator and denominator varying as functions of Ω_m, preserving the H_0-independence of the ratio.[1]

We run χ^2 minimizers over $\{\Omega_m\}$, $\{\Omega_m, w_a\}$, and $\{\Omega_m, w_0, w_a\}$ to quantify each parameter's contribution.

3 Results

3.1 Ω_m Deficit: Confirmed in DR1 and DR2

DR1 (7-bin D_M/r_s, diagonal errors). Applying ξ-normalization to all 7 DESI DR1 bins with self-consistent $r_s(\Omega_m)$ scaling, a 1D χ^2 minimizer yields $\Omega_m = 0.268$ ($\chi^2 = 6.51$ vs. 13.57 for Planck; $\Delta\chi^2 = 7.06$). The matter density deficit is directionally confirmed by DESI DR1 BAO ($\Omega_m = 0.295 \pm 0.010$; DESI Collaboration 2024a) and the S_8 weak lensing tension (Heymans et al., 2021; Amon et al., 2022).

[1] An earlier version of this analysis held r_s fixed at the Planck fiducial while varying Ω_m. This inverts the sign of the Ω_m sensitivity and produced a spurious w_0w_a improvement ($\Delta\chi^2 = 4.74$). With self-consistent $r_s(\Omega_m)$ scaling, the w_0w_a improvement collapses to $\Delta\chi^2 = 0.12$ (Table 2). All results in this paper use the corrected formulation.

DR2 (13-bin $D_M/r_s + D_H/r_s$, full covariance matrix). Using the full DESI DR2 consensus dataset (DESI Collaboration, 2025)—13 measurements with the complete 13×13 covariance matrix from the Cobaya repository (arXiv:2503.14738)—the Ω_m-free fit yields $\Omega_m = 0.290$ ($\chi^2/\text{dof} = 1.75$). This matches the DR2 headline value of 0.289 ± 0.007 exactly, validating the DR1 prediction at the same analysis level. The Planck ΛCDM is rejected at $\chi^2/\text{dof} = 2.41$.

Our DR1 best-fit $\Omega_m = 0.268$ is lower than the DR2/DESI value of 0.290. This discrepancy is expected: the DR1 analysis uses diagonal errors and the Percival approximation for r_s, while DR2 employs the full covariance matrix. The direction of the deficit is robust across all analyses.

3.2 The D_H Dark Energy Signal in DR2

DR1 result. The LRG1 bin at $z = 0.51$ shows a $+1.80\sigma$ residual in the DR1 D_M/r_s analysis. Adding w_0w_a freedom reduces χ^2 by only 0.12 for two additional parameters ($p = 0.94$): no statistical support for dynamical dark energy in the D_M-only dataset.

DR2 result: a third residual pattern. DR2 adds D_H/r_s measurements at each redshift, providing direct sensitivity to $H(z)$ independent of transverse distances. With the full covariance matrix, adding w_0w_a freedom to the DR2 Ω_m-free fit yields $\Delta\chi^2 = 13.13$ ($\chi^2/\text{dof} = 0.79$, best-fit $\Omega_m = 0.333$, $w_0 = -0.710$, $w_a = -1.075$). For two additional parameters this is statistically significant ($p \approx 0.001$). The best-fit w_0 is in the same direction as the DESI Collaboration's combined BAO+CMB+SNe result ($w_0 \approx -0.73$; DESI Collaboration 2025), providing a consistency check on the signal direction.

The largest single-bin D_H/r_s residuals in the Ω_m-free fit are at $z = 0.51$ (LRG1: -1.84σ) and $z = 0.71$ (LRG2: -2.21σ). To test whether these drive the w_0w_a improvement, we perform a leave-one-out jackknife: we refit dropping each of the 13 measurements in turn and record the residual $\Delta\chi^2(w_0w_a)$. The result is shown in Table 1. The signal is *not* a two-bin artifact: removing any single measurement leaves $\Delta\chi^2 \geq 7.4$. The leading contributors are LRG2 (both D_M and D_H; $z = 0.71$), Lyα ($z = 2.33$), and LRG3+ELG1 ($z = 0.93$). The signal is distributed across $z = 0.5$–2.3 in a pattern consistent with a genuine change in the expansion history rather than a single-tracer systematic.

The D_M-only data carry no dark energy sensitivity at this precision; the w_0w_a improvement in the DR1 D_M-only analysis was $\Delta\chi^2 = 0.12$. The emergence of $\Delta\chi^2 = 13.13$ in DR2 is a direct consequence of the added $H(z)$ information from D_H/r_s, not a change in the underlying data or a methodological difference.

4 Discussion

4.1 A Three-Layer Residual Decomposition

Under ξ-normalization, the Hubble tension residuals decompose into three layers with distinct observational signatures, resolved sequentially as data precision increases:

1. **∼90% coordinate artifact** — removed exactly by ξ-normalization; the H_0 factor cancels for all redshift-derived distances. This layer is resolved by the analysis in Martin (2025b) and requires no new data.

Table 1: Leave-one-out jackknife on $\Delta\chi^2(w_0 w_a)$. Each row gives $\Delta\chi^2$ when the listed measurement is excluded. Baseline (all 13 bins): $\Delta\chi^2 = 13.13$. No single bin drives the signal; removing any one leaves $\Delta\chi^2 \geq 7.4$.

Sample	Observable	z	$\Delta\chi^2$
(all 13 bins)	—	—	13.13
LRG2	D_H/r_s	0.706	7.41
Lyα	D_H/r_s	2.330	8.94
LRG2	D_M/r_s	0.706	8.87
LRG3+ELG1	D_M/r_s	0.934	9.38
LRG1	D_H/r_s	0.510	10.98
ELG2	D_M/r_s	1.321	11.09

Table 2: χ^2 goodness-of-fit comparison. Panel A: 7-bin D_M/r_s analysis (DESI DR1, diagonal errors, `bao_wa_minimizer.py`). Panel B: 13-bin $D_M/r_s + D_H/r_s$ analysis (DESI DR2, full 13×13 covariance, `bao_desi_dr2.py`).

Model	Ω_m	w_0	w_a	χ^2	χ^2_ν	Verdict
Panel A: DESI DR1 — 7 bins, D_M/r_s only						
Planck ΛCDM	0.315	-1.0	0.0	13.57	2.26	Rejected
Ω_m free	0.268	-1.0	0.0	6.51	1.09	Good
$\Omega_m + w_0 + w_a$	0.251	-1.101	$+0.360$	6.39	1.60	$\Delta\chi^2 = 0.12$, no support
Panel B: DESI DR2 — 13 bins, $D_M + D_H/r_s$, full covariance						
Planck ΛCDM	0.315	-1.0	0.0	26.5	2.41	Rejected
Ω_m free	0.290	-1.0	0.0	19.2	1.75	Improved
$\Omega_m + w_0 + w_a$	0.333	-0.710	-1.075	8.7	0.79	$\Delta\chi^2 = 13.13$, strong fit improvement

2. **Matter density deficit** — a global $\Omega_m < 0.315$ supported by DR1 ($\Omega_m = 0.295$; DESI Collaboration 2024a), our DR1 ξ-analysis ($\Omega_m = 0.268$), and our DR2 full-covariance analysis ($\Omega_m = 0.290$, consistent with the DR2 headline 0.289 ± 0.007). Independent corroboration comes from S_8/σ_8 weak lensing (Heymans et al., 2021; Amon et al., 2022). This layer is visible in D_M/r_s data alone.

3. **Provisional D_H-driven line-of-sight residual pattern** — a $\Delta\chi^2 = 13.13$ fit improvement from $w_0 w_a$ freedom ($w_0 = -0.710$, $w_a = -1.075$), concentrated at $z = 0.51$–0.71 in the D_H/r_s (line-of-sight) observable. This pattern is *invisible* in D_M-only data and was correctly reported as absent in the DR1 analysis ($\Delta\chi^2 = 0.12$). Its appearance in DR2 is a direct consequence of the added $H(z)$ sensitivity from D_H/r_s. Whether this reflects genuine dark energy evolution or unresolved systematics in the DR2 line-of-sight data requires independent confirmation.

This hierarchical decomposition has a natural methodological interpretation. The coordinate artifact is removed analytically. The Ω_m deficit is resolved by transverse BAO (D_M/r_s). The dark energy signal requires line-of-sight BAO (D_H/r_s) to emerge. Each layer requires progressively higher-dimensional data to resolve, which explains why the dark energy signal was not detectable in the DR1 D_M-only dataset.

4.2 The S_8 Tension as a Matter Density Tie-Breaker

The most compelling independent evidence for a physical Ω_m deficit will arrive with the Euclid DR1 S_8 results (Euclid Collaboration et al., 2022). The S_8 parameter, $\sigma_8(\Omega_m/0.3)^{0.5}$, measures the amplitude of matter clustering and has consistently trended 2–3σ below the Planck ΛCDM prediction in prior weak lensing surveys (Heymans et al., 2021; Amon et al., 2022). Within our ξ-normalized framework, this clustering tension is not a separate crisis but a secondary manifestation of the same Ω_m deficit. A genuine deficit in Ω_m relative to Planck mathematically forces a lower S_8 value. Confirmation of $S_8 < 0.83$ at high significance would constitute strong independent evidence for the matter density hypothesis over a coordinate artifact.

4.3 Reply to the JWST 8σ Objection

The JWST NIRCam result (Riess et al., 2023) confirming $H_0 = 73.17 \pm 1.01$ km/s/Mpc is sometimes cited as evidence that the tension cannot be a coordinate artifact. As shown in Martin (2025b), JWST measures the artifact more precisely within a fixed H_0 frame; it does not test whether that frame and the Planck frame occupy compatible coordinates. The Ω_m deficit identified in this work emerges from the internal geometry of the DESI BAO dataset, which is H_0-independent by construction and therefore orthogonal to the systematic uncertainties of the local distance ladder.

4.4 Limitations and Non-Claims

This analysis has several important limitations. (1) The DR1 χ^2 minimizer uses only D_M/r_s observables with diagonal errors; the DR2 analysis uses the full covariance matrix and both D_M and D_H observables. The DR1 $\Omega_m = 0.268$ and DR2 $\Omega_m = 0.290$ differ partly due to this methodological difference, not a physical discrepancy. (2) The $r_s(\Omega_m)$ scaling from Percival et al. 2007 is an approximation. A sensitivity test varying the power-law exponent α from -0.240 to -0.270 (bracketing the Percival value of -0.255) yields $\Omega_m = 0.288$–0.294, a spread of 0.006 comparable to the DESI measurement uncertainty of 0.007. The qualitative conclusion ($\Omega_m < 0.315$) is robust across all prescriptions tested, including a calibrated Eisenstein & Hu (1998) approximation. A more accurate treatment would vary both $\Omega_b h^2$ and $\Omega_m h^2$ jointly and is expected to bring the central value into closer agreement with the DESI DR2 headline. (3) The DR2 $\Delta\chi^2 = 13.13$ from $w_0 w_a$ is statistically significant within this fit, but the DR2 covariance matrix may carry systematic uncertainties not fully captured here. We report this as a signal for further investigation, not a confirmed detection. (4) The D_H/r_s residuals at $z = 0.51$ and $z = 0.71$ may reflect unresolved systematics in the LRG sample selection rather than dark energy; DR2 has not yet published a definitive systematic budget for these measurements. (5) All conclusions are conditioned on the DESI public data releases; independent confirmation from DESI Year 5, EUCLID DR1, and LSST will be required to elevate any component from signal to detection.

5 Conclusion

Within the ξ-normalized BAO framework, the Hubble tension residuals decompose into three layers. The dominant layer ($\sim$90%) is a coordinate artifact removed analytically

by horizon-normalization. A second layer—a matter density deficit ($\Omega_m \approx 0.290$)—is supported by both the DESI DR1 and DR2 analyses presented here and by independent weak lensing data. A third residual pattern appears only in the DESI DR2 line-of-sight D_H/r_s data: $\Delta\chi^2 = 13.13$ for w_0w_a freedom, concentrated at $z = 0.51$–0.71. This signal was invisible in the DR1 D_M-only dataset ($\Delta\chi^2 = 0.12$); its appearance is a direct consequence of the added $H(z)$ sensitivity from D_H/r_s measurements.

The ξ-normalization framework provides the infrastructure to separate these layers: each requires a different data observable to detect, and each is indicated at the scale warranted by current data quality. The framework predicted the Ω_m deficit direction from DR1 data; the DR2 result is consistent with that prediction and reveals a third residual pattern warranting further investigation. Definitive characterisation of all three layers awaits independent confirmation from DESI Year 5 and Euclid DR1.

Competing Interests

The author holds US Provisional Patent 63/902,536 (Universal Horizon Address, UHA) covering the ξ coordinate infrastructure described in this work (Zenodo DOI: 10.5281/zenodo.17435574). The theoretical results reported here are independent of the patent claims and do not constitute commercial promotion.

Data Availability

All analysis code is publicly available at `https://github.com/abba-01/uha`. Reproduction scripts: `bao_wa_minimizer.py` (DR1, 7-bin D_M/r_s), `bao_desi_dr2.py` (DR2, 13-bin D_M+D_H/r_s, full covariance), `bao_wa_discriminator.py`, `uha_evolution_plot.py`. The DESI DR1 dataset is the public release from DESI Collaboration (2024a). The DESI DR2 covariance matrix is from the Cobaya repository (arXiv:2503.14738; DESI Collaboration 2025). This manuscript is deposited at Zenodo: DOI 10.5281/zenodo.19304441. No proprietary data were used.

References

Amon A., et al., 2022, Phys. Rev. D, 105, 023514, DOI: 10.1103/PhysRevD.105.023514

Brout D., et al., 2022, ApJ, 938, 110, DOI: 10.3847/1538-4357/ac8e04

Chevallier M., Polarski D., 2001, Int. J. Mod. Phys. D, 10, 213, DOI: 10.1142/S0218271801000822

DESI Collaboration, 2024a, arXiv:2404.08633

DESI Collaboration, 2024b, arXiv:2404.08056

DESI Collaboration, 2025, arXiv:2503.14738

Euclid Collaboration, Scaramella R., et al., 2022, A&A, 662, A112, DOI: 10.1051/0004-6361/202141938

Freedman W. L., 2024, ApJ, 961, 32, DOI: 10.3847/1538-4357/ad1bac

Heymans C., et al., 2021, A&A, 646, A140, DOI: 10.1051/0004-6361/202014214

Linder E. V., 2003, Phys. Rev. Lett., 90, 091301, DOI: 10.1103/PhysRevLett.90.091301

Martin E. D., 2025a, Universal Horizon Address, US Patent 63/902,536, DOI: 10.5281/zenodo.17435574

Martin E. D., 2025b, The Hubble Tension as a Measurement Artifact, DOI: 10.5281/zenodo.19211662

Percival W. J., Cole S., Eisenstein D. J., Nichol R. C., Peacock J. A., Pope A. C., Szalay A. S., 2007, MNRAS, 381, 1053, DOI: 10.1111/j.1365-2966.2007.12268.x

Planck Collaboration, 2020, A&A, 641, A6, DOI: 10.1051/0004-6361/201833910

Poulin V., et al., 2019, Phys. Rev. Lett., 122, 221301, DOI: 10.1103/PhysRevLett.122.221301

Riess A. G., et al., 2022, ApJL, 934, L7, DOI: 10.3847/2041-8213/ac5c5b

Riess A. G., et al., 2023, ApJL, 956, L18, DOI: 10.3847/2041-8213/acf769